ROOPESH S

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EDUCATION

Indian Institute of Information Technology, Lucknow **Nov 2022 – June 2026**
B.Tech, Computer Science and Business *CGPA: 8.5*

TECHNICAL SKILLS

Programming Languages: Java, C++, Python, Rust, TypeScript, JavaScript
AI & Agentic Workflows: LangGraph, LangChain, AI Agents, LLM Orchestration, Playwright
Backend & Frontend: Node.js, ExpressJS, Java (Spring Boot), ReactJS, Redux, Next.js
Databases & Cache: MongoDB, MySQL, Redis
Tools & Platforms: Linux, Git, CI/CD, KONG Gateway, Kibana

WORK EXPERIENCE

Microsoft **Noida, India**
Software Engineer *Aug 2026 - Present*

- Built **TestIQ** an AI-powered natural-language testing framework orchestrated with **LangGraph**, **LangChain**, and **Playwright** that runs plain-English test cases daily on **CI/CD pipelines** across 3 productivity apps reducing manual testing effort by **80%**.
- Developed **BugPilot**, an autonomous **AI agent** that reproduces, fixes, and validates UI bugs; analyzed **600+ bugs**, delivered **450+ merged PRs**, and added video analysis to detect flicker and jitter.

ACKO General Insurance **Bangalore, India**
Software Engineering Intern *Dec 2025 - July 2026*

- Extended existing claims workflow by integrating **General Insurance** and **Life Insurance** and structured frontend state using **Redux** for better maintainability.
- Implemented **rate limiting** at both **application level** and **KONG Gateway** to prevent API abuse and improve system reliability.
- Developed **KYC and fraud detection flow** by integrating **Spectra API**, improving classification accuracy by **27%**.

Microsoft **Noida, India**
Software Engineering Intern *June 2025 - Aug 2025*

- Built **Visualize as Table** feature in PowerPoint Copilot web, enabling structured table generation from rewritten content and improving document authoring workflows.
- Enhanced **backend services** supporting Copilot rewrite flows by fixing production issues and improving request handling, contributing to a **3.8% increase** in rewrite feature adoption.

PROJECTS

ConnectExpress
Next.js, Node.js, Express.js, MongoDB, Python, C++ *Oct 2024 - Nov 2024*

- Built a railway route finder using a **two-pass connection-matching algorithm** in **C++ (N-API)** compiled as a Node.js native addon, supporting multi-leg journey planning with layover validation.
- Developed a delay prediction model on historical and weather data achieving ∼**85% accuracy**; integrated **Redis caching** reducing response time by ∼**40%** under peak load.

OPEN SOURCE

- Contributed **50+ pull requests** across repositories including NVIDIA DSX, RTK-AI, Carbon Language, LibreFang, OpenWork, and PinchTab.
- **NVIDIA DSX (Rust):** Shipped CLI features and API bug fixes for NVIDIA's hardware lifecycle management and multitenant networking controller.
- **RTK-AI (Rust):** Resolved CLI proxy and agent hook issues for a developer tool that reduces LLM token consumption by **60–90%**.

ACHIEVEMENTS

- Rated Specialist on **CodeForces** (Max Rating: **1518**) and 4-Star on **CodeChef** (Max Rating: **1852**).
- Selected for **Amazon ML Summer School** in consecutive years (**2024, 2025**).
- Winner at **SuperMove Tour DAppathon** – Developed **Meta EDU**, an AR/VR project on the Aptos blockchain.
- Runner-up at **HackOdisha 3.0** and Second Runner-up at **Hack-O-Fiesta V4.0** (IIIT Lucknow, Blockchain Track).
- Finalist at **Dapp-Thon** (top teams among **1,000+** global participants) and Semi-Finalist at **Flipkart Grid 6.0**.